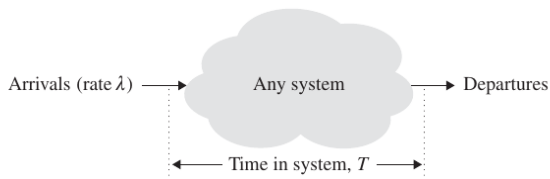


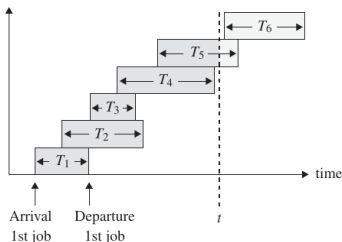
Little's law



$$E[N] = \lambda E[T]$$

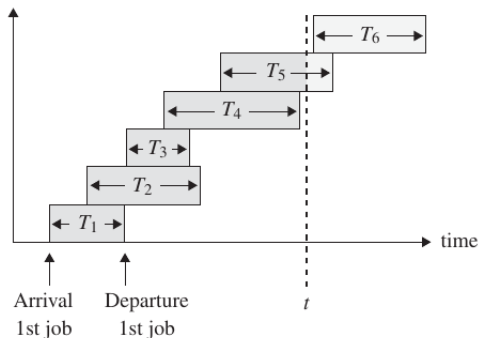
$$\left[\begin{array}{l} \text{Avg. number of} \\ \text{people in Ikea} \end{array} \right] = \lambda \left[\begin{array}{l} \text{Avg. time spent} \\ \text{per customer in Ikea} \end{array} \right]$$

Little's law: Intuitive ensemble average proof



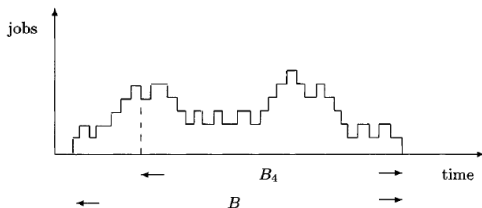
- ▶ Let $F(\cdot)$ denote the CDF of T_i .
- ▶ At time t , consider those in the system have arrived before t and are staying beyond t .
- ▶ At $(t - x, t - x + dx)$, jobs arrive with probability λdx .
- ▶ Each such job will stay beyond t with probability $1 - F(x)$
- ▶ $E[N(t)] = \int_0^t [1 - F(x)] \lambda dx$.
- ▶ $L = \lim_{t \rightarrow \infty} E[N(t)] = \int_0^\infty [1 - F(x)] \lambda dx$.

Little's law: Intuitive time average proof



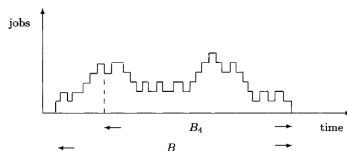
- ▶ $N(t) = A(t) - D(t)$
- ▶ $\sum_{n=1}^{D(t)} T_n \leq \int_0^t N(t) \leq \sum_{n=1}^{A(t)} T_n$
- ▶ $\frac{D(t)}{t} \frac{\sum_{n=1}^{D(t)} T_n}{D(t)} \leq \frac{\int_0^t N(t)}{t} \leq \frac{A(t)}{t} \frac{\sum_{n=1}^{A(t)} T_n}{A(t)}$
- ▶ What is $\lim_{t \rightarrow \infty} \frac{D(t)}{t}$?

Busy cycles of a work-conserving system



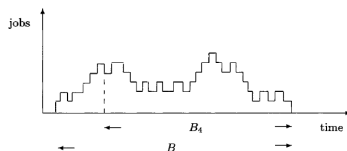
- ▶ When the system is work conserving, the system oscillates between busy periods and idle periods.
- ▶ Busy period + idle period = Busy cycle
- ▶ The start of a busy cycle constitutes a renewal point.
- ▶ The interarrival times between busy cycles are i.i.d
- ▶ We can therefore also use renewal-reward theorems to calculate L , W and even $\lim D(t)/t$.

Busy cycles of a work-conserving system



- ▶ Consider $\lim_{t \rightarrow \infty} D(t)/t$.
- ▶ Let C_i denote the length of the i^{th} cycle.
- ▶ Let N_i denote the number of jobs served in i^{th} cycle.
- ▶ Assume that every departure earns a reward of 1 unit. Then in i^{th} cycle, the reward earned is N_i .

Busy cycles of a work-conserving system



- ▶ From renewal reward theorem, we have $\frac{D(t)}{t} \rightarrow \frac{E[N]}{E[C]}$
- ▶ Using Wald's lemma, we can show that $E[C] = \frac{E[N]}{\lambda}$.
- ▶ This implies that $\frac{D(t)}{t} \rightarrow \lambda$.

(See Thm 3.62 and Section 3.6.1 of Sheldon Ross)

Consequences of Little's law

- ▶ let N_q denote the mean number of waiting jobs in a queueing system at stationarity.
- ▶ Similarly N denotes the mean number of jobs in the system.
- ▶ $N_s = N - N_q$ denotes the mean number of jobs receiving service.
- ▶ W_q denotes the mean time spent by any job waiting for service while S denotes the mean sojourn time.
- ▶ $S - W_q$ denotes the mean service time.

Consequences of Little's law – $M/M/1$

► $N_q = ?$

► $N = ?$

► $N_s = N - N_q = ?$

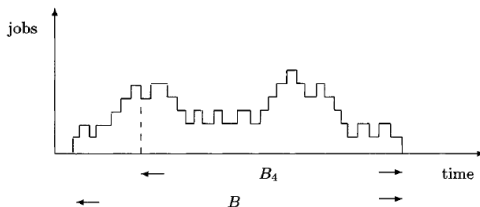
► $W_q = \frac{\rho}{\mu - \lambda}$

► $S = \frac{1}{\mu - \lambda}$

► $S - W_q = ?$

Exercise:- Identify these for $M/M/1/K$, $M/M/K/K$ and $M/M/K/\infty$

Busy period analysis for $M/M/1$



- ▶ What is the mean length of busy period, i.e., $E[B]$?
- ▶ What is the probability that the server is busy? ($1 - \pi_0 = \frac{\lambda}{\mu}$)
- ▶ The time average that the server is busy is $\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t 1_{\{N(t) > 0\}} dt$.
- ▶ Using RR theorem, this is equal to $\frac{E[B]}{E[B] + \frac{1}{\lambda}}$
- ▶ Equating the two averages give us $E[B] = \frac{1}{\mu - \lambda}$.
- ▶ Mean number of jobs served in a busy period $n_B = E[B] / \frac{1}{\mu}$.